

An Analysis of Artificial neural network

Artificial neural network

$$a^{(l)} = \sigma(W^{(l)}a^{(l-1)} + b^{(l)})$$

Artificial neural network -- Part 1

Computational power The multilayer perceptron is a universal function approximator, as proven by the universal approximation theorem. However, the proof does not specify the number of neurons required, the network topology, the weights or the learning parameters. A recurrent architecture with rational-valued weights (as opposed to full precision real number-valued weights) has the power of a universal Turing machine, using a finite number of neurons and linear connections. Further, the use of irrational values for weights results in a machine with super-Turing power.

Artificial neural network -- Part 2

Natural language processing In natural language processing, NNs are used for tasks such as text classification, sentiment analysis, machine translation, to answer free-form questions, act as chatbots, and to summarize and analyze texts. This has implications for automated customer service, content moderation, and language understanding technologies.

Artificial neural network -- Part 3

Cybersecurity Neural networks are widely applied in cybersecurity for anomaly detection, malware classification, and intrusion detection. By learning patterns of normal system or network behavior, NNs can identify deviations that indicate malicious activity.

Artificial neural network -- Part 4

While it is possible to assess a loss function ad hoc, typically it must exhibit desirable properties such as convexity, differentiability, and robustness. In a probabilistic model, the model's posterior probability can be used as an inverse cost (higher values are better).

Artificial neural network -- Part 5

Error-based monitoring: comparing current predictions against ground-truth labels. This approach directly quantifies predictive performance but may be impractical when labels are delayed or impractical to obtain. **Data distribution monitoring:** detecting changes in the input data distribution using statistical tests, divergence measures, or density-ratio estimation. **Representation monitoring:** tracking the distribution of internal embeddings or

hidden-layer features. Shifts in the latent representation can indicate nonstationarity even when labels are unavailable. Statistical methods such as statistical process control charts have been adapted for this purpose.

Artificial neural network -- Part 6

In machine learning, a neural network (NN) or neural net, is a computational model inspired by the structure and functions of biological neural networks. A neural network consists of connected units or nodes called artificial neurons, which loosely model the neurons in the brain. Artificial neuron models that mimic biological neurons more closely have also been recently investigated and shown to significantly improve performance. These are connected by edges, which model the synapses in the brain. Each artificial neuron receives signals from connected neurons, then processes them and sends a signal to other connected neurons. The "signal" is a real number, and the output of each neuron is computed by some non-linear function of the totality of its inputs, called the activation function. The strength of the signal at each connection is determined by a weight, which adjusts as part of the training process. Groups of neurons are aggregated into layers. Each layer performs a transformation on its inputs. Signals travel from the first layer (the input layer) to the last layer (the output layer), typically passing through multiple intermediate layers (hidden layers). A network is typically called a deep neural network if it has at least two hidden layers. Deep neural networks are capable of learning sophisticated hierarchical representations. Training neural networks is a compute-intensive process, accelerated by the use of graphics processing units (GPUs), and large datasets. Architectural innovations such as convolutional neural networks (CNNs) significantly improved performance in computer vision tasks, while recurrent neural networks (RNNs) enabled modeling of sequential data such as speech and time-series information. Transformer architectures introduced attention mechanisms that allow neural networks to model long-range dependencies in data and have the basis of large language models. Artificial neural networks are used for a myriad of tasks including chatbots, large-scale text, image, and video generation, and robotics.

Artificial neural network -- Part 7

Lack of interpretability NNs are "black box" systems that have complicated understanding of their decision-making processes. NNs are further vulnerable to adversarial examples ("poison"), that can cause incorrect predictions. These concerns have led to increased research in explainable artificial intelligence (XAI), robust machine learning, and hybrid AI approaches that combine neural learning with symbolic reasoning. Advocates of hybrid

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models also say that such a mixture can better capture the mechanisms of the human mind. Progress has included local vs. non-local learning and shallow vs. deep architectures. Analyzing human thought (a biological neural network) has also proven difficult.

Artificial neural network -- Part 8

Unsupervised learning In unsupervised learning, input data is given along with the cost function for the data x and the output, without an "answer sheet". The cost function is dependent on the task (the model domain) and reflects a priori assumptions (implicit model properties, its parameters and the observed variables). For example, the model $f(x) = a$ treats a as a constant and the cost $C = E[(x - f(x))^2]$. Minimizing this cost produces a value of a that is equal to the mean of the data. The cost function can be much more complicated. Its form depends on the application: for example, in compression it could be related to the mutual information between x and $f(x)$, whereas in statistical modeling, it could be related to the posterior probability (in both examples, those quantities are maximized). Unsupervised learning is typically applied to estimation problems; applications include clustering, the estimation of statistical distributions, compression, and filtering.

Artificial neural network -- Part 9

Stochastic neural network Stochastic neural networks originating from Sherrington–Kirkpatrick models are a type of neural network built by introducing random variations into the network, either by giving neurons stochastic transfer functions, or by giving them stochastic weights. This makes them useful tools for optimization problems, since the random fluctuations help the network escape from local minima. Stochastic neural networks trained using a Bayesian approach are known as Bayesian neural networks.

Artificial neural network -- Part 10

Image processing NNs are employed in computer vision tasks such as image classification, object and facial recognition, and image segmentation. They have been applied to automated surveillance medical imaging for diagnosis.

Artificial neural network -- Part 11

Function approximation, or regression analysis, (including time series prediction, fitness approximation, and modeling) Data processing (including filtering, clustering, blind source separation, and compression) Nonlinear system identification and control (including vehicle control, trajectory prediction, adaptive control, process control, and natural resource management) Pattern recognition (including radar systems, face identification, signal classification, novelty detection, 3D reconstruction, object recognition, and sequential decision making) Sequence recognition (including gesture, speech, and handwritten and printed text recognition) Sensor data analysis (including image analysis) Robotics (including directing manipulators and prostheses) Data mining (including knowledge discovery in databases) Finance (such as ex-ante models for specific financial long-run forecasts and artificial financial markets) Quantum chemistry General game playing Generative AI Data visualization Machine translation Social network filtering E-mail spam filtering Medical diagnosis Disaster response NNs have been used to diagnose cancers and to distinguish highly invasive cancer cell lines from less invasive lines using only cell shape data. NNs have been used to accelerate reliability analysis of infrastructures subject to natural disasters and to predict settling in building foundations. They are used to mitigate flooding by modelling rainfall-runoff. NNs have been used for building black-box models in geoscience: hydrology, ocean modelling and coastal engineering, and geomorphology. NNs have been employed in cybersecurity, with the objective to discriminate between legitimate and malicious activities. For example, machine learning has been used for classifying Android malware, for identifying domains belonging to threat actors and for detecting URLs posing a security risk. Research is underway on penetration testing, for detecting botnets, credit cards frauds, and network intrusions. NNs have been proposed as a tool to solve partial differential equations in physics and simulate the properties of many-body open quantum systems. In brain research NNs have studied short-term behavior of individual neurons, the dynamics of neural circuitry arise from interactions between individual neurons and how behavior can arise from abstract neural modules that represent complete subsystems. Studies considered long-and short-term plasticity of neural systems and their relation to learning and memory from the individual neuron to the system level. NNs show promise in profiling a user's interests from photos and discovering new stable materials by efficiently predicting the total energy of crystals.